



BENTELER Steel/Tube
1 Benteler Drive|Shreveport|LA 71115|USA
Tel.: +1 318-216-4149
qms.shreveport@benteler.com

INSPECTION CERTIFICATE EN 10204-3.1
EN 10204:2005-01

Pipe Distributors, Inc.
PO Box 23237
HOUSTON TX 77028
USA

Document No.: 465-1360989/001/P
Purchase Order No.: 6652
Benteler Order No.: 48002815
Dispatch Note No.:
Product: SEAMLESS STEEL TUBES
Inspection No.:
Manufacturer: BSTMC Shreveport USA
Manufacturer's brand:
Stamp of the inspection representative: MI
Steelmaking process: ELECTRIC FURNACE

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Terms of delivery: ASTM-A 53-2018, ASME SA-53, ASME BPVC.II.A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, API Specification 5L
Forty-Sixth Edition April 2018 Incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175/ISO 15156-1: 2015, ANSI/NACE MR0175/ISO 15156-2:
2015, ANSI/NACE MR0103/ISO 17945:2015
Dimensions-tolerances: ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, ASTM-A 53-2018, ASME SA-53, ASME BPVC.II.A-2021, API Specification 5L
Forty-Sixth Edition April 2018 Incl. Errata 1, May 2018
Steel grade: GRADE B, GRADE C, GRADE BN, GRADE X42N
Delivery condition: N

Product marking: FS: BENTELER BENTELER-Sign API 5L-1072 API-Sign Monthly Year Conditions Dimension: O.D. * t inch Steel Grade
PSL2 SMLS Heat No. Tube length feet Test Pressure PSI Production No. Customer's material no.
AEZ = Etching ink marking FK = color marking FS = color jet printer FSD = color jet printer FSI = color jet printer inside UK = Laser marking PXE = laser marking PS = ink jet spray marking

Item	Dimensions	Heat No.	Test pressure PSI	sec
0002	2 NPS - SCH 80 35.000 FT - 45.000 FT	314373	2970	5

Country of melt and pour
Melted & Manufactured in USA
Nucon, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	314373	0.160	0.247	0.98	0.011	0.003	0.10	0.02	0.08	0.20	0.004	0.002	0.002	0.0003

1. Formula: $CE = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0.43 \%$
2. Formula: $Cr + Cu + Mo + Ni + V \leq 1.00 \%$
3. Formula: $Nb + V \leq 0.06 \%$
4. Formula: $Nb + Ti + V \leq 0.15 \%$



Heat 314373

Desc Pipe SMLS ASTM A106-B/C, A53-B, API 5L B/X42,2.375 x .218 - 2" XH SRL



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Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	314373	0.367	0.404	0.006	0.008

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	314373	0.170	0.260	0.99	0.010	0.003	0.10	0.02	0.08	0.20	0.004	0.001	0.003	0.0002
0002	314373	0.170	0.260	0.99	0.012	0.004	0.10	0.02	0.08	0.20	0.003	0.001	0.003	0.0003

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr+Mo+V)/5) + ((Cu+Ni)/15) \leq 0.43 \%$
2. Formula: $C + Cu + Mo + Ni + V \leq 1.00 \%$
3. Formula: $Nb + V \leq 0.06 \%$
4. Formula: $Nb + Ti + V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	314373	0.378	0.404	0.005	0.008
0002	314373	0.378	0.403	0.004	0.007

Test results

The tubes have been submitted to a leak tightness test by: Hydrostatic test acc. to API 5L, Test pressure/Time-record, the test records/charts shall be available for examination by the inspector

The tubes are non destructive tested: ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM E 309; ET - Test drilled hole: 0.125 IN.; ET - Notch shape/hole: drilled hole

Visual inspection: PASSED Bending test: PASSED Dimensions examination: PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	RT 0.5/Rm
			IN	RT 0.5 % PSI	Rm PSI	A2" %		MAX 0.93
0002	000011	314373	0.748 X 0.219	50618	76290	33.00	19.00	0.66
0002	001562	314373	0.749 X 0.221	51779	76145	34.00	19.00	0.68
0002	003111	314373	0.749 X 0.220	51488	76435	32.00	19.00	0.67

Heat 314373

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Elongation formula1. Formula $AF=625000 \cdot AXC^{0,2} / U^{0,9}$ Protection

outside surface varnished

Item

Heat No.

Production No.

0002

314373

15026644

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156; HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945; HRC max 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Data converted from SI units, No weld repair has been carried out;
Certificate-Remark: LONGHORN

Plant 465, 1 BENTELER DRIVE, LA 71115 Shreveport, 11/22/2022, TEL.: 0318-216-4149

Inspection representative

PIDICHETTI / ML

We certify that the supplied product has been processed, sampled, tested, and inspected in accordance with the agreed specifications and/or references mentioned herein and meet the requirements for the grade indicated. This document was prepared by means of electronic data processing and is valid without signature.



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Delivery condition:

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PSL2 SMLS Heat No. Tube length feet Test Pressure PSI Production No. Customer's material no.

AEZ = Etching ink marking PK = color marking FS = point stenciling FSD = Color jet printer PSI = Color jet printer inside LX = Laser marking PKE = bag marking PS = die stamp TS = Ink jet spray marking

Item Dimensions

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Nucor, Jewett, TX, US

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PASSED

The tubes are non destructive tested:

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PASSED

Visual inspection:

PASSED

Bending test

PASSED

Dimensions examination:

PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0.5/Rm
			IN	RT 0.5 % PSI	Rm PSI	A2" %		
				42100-71800	70000-95000			MAX 0.93
0002	000011	314373	0.748 X 0.219	50618	76290	33.00	19.00	0.66
0002	001562	314373	0.749 X 0.221	51779	76145	34.00	19.00	0.68
0002	003111	314373	0.749 X 0.220	51488	76435	32.00	19.00	0.67